

AWS EC2 Hands-on Guide

Instance Connect, Instance Roles, Purchasing Options, Stop/Hibernate, Spot Instances and Spot Fleet

1. EC2 Instance Connect

A way to open an SSH terminal to your instance **from the browser**, with no key file to manage. When you click Connect, AWS pushes a **temporary public key** (valid for about 60 seconds) to the instance, and the browser session uses the matching private key. Access is controlled by IAM permission (`ec2-instance-connect:SendSSHPublicKey`), so you can grant or remove it per user.

Hands-on

- 1 EC2 > Instances > select a **running** instance > **Connect**.
- 2 Choose the **EC2 Instance Connect** tab. The user name is pre-filled (`ec2-user` for Amazon Linux, `ubuntu` for Ubuntu).
- 3 Click **Connect**. A browser terminal opens. Try `whoami` and `hostname`.

Requirements and common problems

Requirement	Detail
AMI	Amazon Linux 2 / 2023 and recent Ubuntu include the EC2 Instance Connect software. Not for Windows.
Security group	Port 22 must be open to the EC2 Instance Connect service range for your Region (or 0.0.0.0/0 for learning only). If 22 is closed, the connection times out.
Network	The instance needs a public IPv4 address for the browser method. Private instances need an EC2 Instance Connect Endpoint.
IAM	Your IAM user needs permission to push the key. Admin users already have it.
Key pair	Not needed. It works even if the instance has no key pair or you lost the file.

Instance Connect vs normal SSH: normal SSH needs your own key file and a client (PuTTY, ssh). Instance Connect needs neither, but still needs port 22 reachable.

2. EC2 Instance Roles Demo

Goal: show that an instance can call AWS APIs with **no access keys**, using an IAM role. (A deeper S3 version of this lab is in the earlier guide.)

Part A: Before the role

Connect with Instance Connect and run:

```
aws iam list-users
# Unable to locate credentials. You can configure credentials by running "aws configure".
```

Do not run `aws configure`. That stores access keys on the server, which is exactly what roles avoid.

Part B: Create and attach a role

- 1 IAM > Roles > **Create role**. Trusted entity: **AWS service**, use case **EC2**.
- 2 Attach the managed policy **IAMReadOnlyAccess**. Name the role DemoRoleForEC2.
- 3 EC2 > select the instance > **Actions > Security > Modify IAM role** > choose DemoRoleForEC2 > Update IAM role.

Part C: After the role

```
aws sts get-caller-identity
# Arn: arn:aws:sts::<account>:assumed-role/DemoRoleForEC2/i-xxxxxxx

aws iam list-users          # now works (read-only)
aws iam create-user --user-name x # AccessDenied (read-only role)
```

If the first call still fails, wait a minute and retry, because role changes take a short time to reach the instance. The role has two parts: a **trust policy** (EC2 may assume it) and a **permissions policy** (what it may do). Credentials are temporary and rotate automatically.

3. EC2 Purchasing Options

Think of renting a hotel: pay nightly with no promise, or commit for a year for a discount, or grab leftover rooms cheaply and risk being asked to leave.

Option	How it works	Discount*	Best for
On-Demand	Pay per second (Linux/Windows) or per hour. No commitment.	None (baseline)	Short, unpredictable or first-time workloads, learning
Reserved Instances	Commit to a specific instance configuration for 1 or 3 years. Standard or Convertible (can change attributes).	Up to ~72%	Steady, predictable workloads such as a database
Savings Plans	Commit to a spend per hour (\$/hour) for 1 or 3 years. Compute plans are the most flexible; EC2 Instance plans are tied to a family and Region.	Up to ~72%	Steady usage where you may change instance types
Spot Instances	Use spare capacity at a variable price. AWS can reclaim the instance with a 2-minute warning.	Up to ~90%	Batch jobs, data analysis, CI/CD, anything that tolerates interruption
Dedicated Hosts	A whole physical server for you. Gives visibility of sockets and cores for licence rules. On-Demand or reserved.	Highest cost	Bring-your-own-licence, strict compliance
Dedicated Instances	Instances on hardware used only by your account. No control over placement.	Premium over shared	Compliance needing isolated hardware
Capacity Reservations	Reserve capacity in one Availability Zone for any length of time. Billed at On-Demand rate whether you use it or not.	None alone (combine with RIs / Savings Plans)	Must-have capacity for an event or critical launch

*Discounts are approximate maximums versus On-Demand and vary by instance type, Region and term. Check the current AWS pricing pages.

Quick chooser: unpredictable or learning = On-Demand. Steady 24/7 = Savings Plan or Reserved.
 Interruption-tolerant = Spot. Licence or compliance on physical hardware = Dedicated Host.
 Must-have capacity = Capacity Reservation.

4. Stopping Instances

Action	What happens	You pay for
Stop	Instance powers off. EBS volumes and data stay. Public IP is released.	EBS storage (and any Elastic IP)
Start	Boots again with a new public IP (the private IP stays).	Compute again
Hibernate	RAM contents saved to the encrypted root volume. Resumes with processes still running. Needs supported types and an encrypted root volume.	EBS storage
Terminate	Instance is deleted. Root volume is deleted by default.	Nothing for the instance (extra volumes may remain)

Hands-on: select a running instance > Instance state > **Stop**. Note the public IPv4 disappears. Then Instance state > **Start** and compare the new address. Because you are not charged compute while stopped, stopping is the easiest way to save money between study sessions.

5. Spot Instances and Spot Fleet

Spot Instances use unused EC2 capacity at a big discount. You can set a maximum price you will pay (by default it is capped at the On-Demand price). When the Spot price rises above your limit, or AWS needs the capacity back, the instance is **interrupted** with a 2-minute warning.

Spot request types

Type	Behaviour after interruption
One-time	Request ends once the instance launches. If it is interrupted, nothing relaunches.
Persistent	Request stays active and launches a replacement whenever capacity and price allow. Interruption behaviour can be terminate, stop or hibernate.

Cancel order matters: cancel the Spot **request** first, then terminate the instances. If you only terminate the instance, a persistent request launches a new one. (Cancelling the request does not by itself terminate running instances, unless you tick the terminate option.)

Spot Fleet

A **Spot Fleet** is a collection of Spot Instances (and optionally On-Demand) that tries to reach a **target capacity** across several instance types and Availability Zones, called **pools**. Mixing pools lowers the chance that all your capacity is interrupted at once.

Allocation strategy	Idea
lowestPrice	Take the cheapest pool. Cheapest but riskier for interruption.
diversified	Spread across all pools. Good for availability.
capacityOptimized	Pick pools with the most spare capacity, so fewer interruptions.

Allocation strategy	Idea
priceCapacityOptimized	Balance low price and high capacity. Recommended for most workloads.

Note: AWS now also offers **EC2 Fleet** and **Auto Scaling groups with mixed instances**, which do the same job with more features. Spot Fleet is still a common exam and course topic.

Hands-on: look at Spot first, then request one

- 1 EC2 > **Spot Requests**. Click **Pricing history** and compare a few instance types and Availability Zones. Notice prices move but are far below On-Demand.
- 2 Click **Request Spot Instances**. Choose an Amazon Linux AMI, instance type t3.micro, and your existing key pair and security group.
- 3 Set **target capacity to 1**. Allocation strategy: **Price capacity optimized**. Leave **Maintain target capacity** off for this test, so it does not relaunch replacements.
- 4 Submit. Open Spot Requests until the status is **fulfilled**, then check Instances for the new Spot instance.
- 5 **Cleanup, in this order**: select the request > Actions > **Cancel request** and tick the option to terminate the instances. Confirm in Instances that it shows terminated.

Cost safety: a small Spot instance costs a fraction of a cent to a few cents per hour, but a forgotten persistent or maintained request keeps launching instances. Always cancel the request.

6. Launch Types: What to Do and What Not To Do

Option	Hands-on?	Why
On-Demand	Yes	Default launch. Safe and cheap on Free Tier eligible types.
Spot instance / Spot Fleet	Yes , with cleanup	Small cost, as long as you cancel the request.
Reserved Instances / Savings Plans	Look only	Open the purchase screens to see the options, but do not buy. They are 1 to 3 year commitments you cannot easily undo.
Dedicated Hosts	Do not launch	Billed for the whole host, can be expensive.
Capacity Reservations	Do not create	Billed at the full On-Demand rate whether used or not.

To explore costs safely, use the **AWS Pricing Calculator** (calculator.aws) to compare On-Demand, Savings Plans and Spot for the same instance.

7. Troubleshooting SSH from Windows

Error	Cause and fix
PuTTY: Unable to use certificate file ... key type 'ssh-rsa' is not a certificate	The .ppk was put in the optional Certificate box. Leave it empty. Put the .ppk only under Connection > SSH > Auth > Credentials > Private key file , then save the session.
PuTTY: No supported authentication methods available (server sent: publickey,...)	The server rejected the key. Usually it is not the key pair the instance was launched with. Check Key pair assigned at launch on the Details tab. A key pair cannot be changed later.
ssh: WARNING: UNPROTECTED PRIVATE KEY FILE / bad permissions	Other Windows accounts can read the .pem. Run the three commands below, then connect again.
Connection timed out	Security group does not allow port 22 from your IP, or the public IP changed after stop/start.

Fix .pem permissions in PowerShell (from the folder containing the key):

```
icacls .\key.pem /inheritance:r
icacls .\key.pem /remove "OTHER-ACCOUNT-NAME-SHOWN-IN-THE-ERROR"
icacls .\key.pem /grant:r "${env:USERNAME}:R"
ssh -i .\key.pem ec2-user@<public-ip>
```

Never commit .pem or .ppk files to GitHub.

8. Cleanup Checklist

- Cancel Spot requests (and terminate their instances).
- Stop or terminate practice instances.
- Detach and delete the DemoRoleForEC2 role if no longer needed.
- Release unused Elastic IPs; delete unattached volumes and old snapshots.
- Keep a billing budget alert active.

9. Check Yourself

Question	Answer
Why can Instance Connect work without a key pair?	AWS pushes a temporary public key for about 60 seconds, controlled by IAM permission.
Which port must be reachable for Instance Connect?	Port 22 (SSH).
Why use an instance role instead of aws configure on the instance?	Role credentials are temporary and rotate automatically, and no secret is stored on the server.
Cheapest option for a job that can be interrupted?	Spot Instances (up to about 90% off).
Which option is billed even when unused?	Capacity Reservations (and Dedicated Hosts, because you pay for the host).
What do you cancel first for a persistent Spot request?	The Spot request, then the instances.
What does stopping an instance still cost?	EBS storage (and any Elastic IP), but not compute.